Generalized Linear Phase

from Oppenheim/Schafer, Section 5.7.2

A system whose frequency response has the form

$$H(e^{j\omega}) = A(e^{j\omega})\, e^{-j\alpha\omega + j\beta}$$

where α, β are constants and $A(e^{j\omega})$ is a real (possibly bipolar) function of ω, is called a generalized linear phase system.

Here the linear phase is of the general form

$$\arg[H(e^{j\omega})] = \beta - \alpha\omega \qquad 0 < \omega < \pi$$

and thus

$$\tau_g(\omega) = -\frac{d}{d\omega}\{\arg[H(e^{j\omega})]\} = \alpha$$

We can express $H(e^{j\omega})$ in two ways as follows:

(1) $H(e^{j\omega}) = A(e^{j\omega})\, e^{j(\beta-\alpha\omega)}$

$$= A(e^{j\omega})\cos(\beta-\alpha\omega) + jA(e^{j\omega})\sin(\beta-\alpha\omega)$$

(2) $H(e^{j\omega}) = \sum_n h[n] e^{-j\omega n}$

Note that $h[n]$ is assumed to be real here

$$= \sum_n h[n]\cos(\omega n) - j\sum_n h[n]\sin(\omega n)$$

Now the tangent of the phase angle of $H(e^{j\omega})$ can be written as

$$\tan(\beta-\alpha\omega) = \frac{\sin(\beta-\alpha\omega)}{\cos(\beta-\alpha\omega)} = \frac{-\sum_n h[n]\sin(\omega n)}{\sum_n h[n]\cos(\omega n)}$$

Cross multiplying yields

$$\sum_n h[n]\cos(\omega n)\sin(\beta-\alpha\omega) = -\sum_n h[n]\sin(\omega n)\cos(\beta-\alpha\omega)$$

Thus

$$\sum_n h[n]\Big(\underbrace{\sin(\beta-\alpha\omega)\cos(\omega n)+\cos(\beta-\alpha\omega)\sin(\omega n)}_{=\sin(\beta-\alpha\omega+\omega n)}\Big)=0$$

and

$$\sum_n h[n]\sin(\beta+(n-\alpha)\omega)=0 \quad \text{for all } \omega \qquad (*)$$

is a necessary condition on $h[n]$, α, and β for a system $H(e^{j\omega})=\sum_n h[n]e^{-j\omega n}$ with real-valued $h[n]$ to have a constant group delay. It is not a sufficient condition, however, and does not tell us how to find $h[n]$, α, β in general.

A class of examples of generalized linear-phase systems are those for which

$$\beta = 0 \text{ or } \pi$$

$$2\alpha = M = \text{an integer}$$

$$h[2\alpha-n]=h[n]$$

Note: M is the filter order
N = M+1 is the number of filter taps

With $\beta = 0$ or π equation $(*)$ becomes

$$\sum_n h[n]\sin((n-\alpha)\omega)\overset{!}{=}0 \quad \text{for all } \omega$$

Setting $2\alpha-n\leftarrow n$, $\sum_n h[2\alpha-n]\underbrace{\sin((2\alpha-n-\alpha)\omega)}_{=\sin((\alpha-n)\omega)=-\sin((n-\alpha)\omega)}\overset{!}{=}0$

and thus the $\sin((n-\alpha)\omega)$ terms cancel for each $h[n]$, $h[2\alpha-n]$ pair since $h[n]=h[2\alpha-n]$

Another class of examples of generalized linear-phase systems are those for which

$$\beta = \pi/2 \text{ or } 3\pi/2$$
$$2\alpha = M = \text{an integer}$$
$$h[2\alpha - n] = -h[n]$$

With $\beta = \pi/2$ or $3\pi/2$ equation (*) becomes

$$\sum_n h[n] \cos((n-\alpha)\omega) \stackrel{!}{=} 0 \quad \text{for all } \omega$$

Setting $2\alpha - n \leftarrow n$ we get

$$\sum_n h[2\alpha - n] \underbrace{\cos((2\alpha - n - \alpha)\omega)}_{= \cos((\alpha - n)\omega) = \cos((n-\alpha)\omega)} \stackrel{!}{=} 0$$

Thus

$$\sum_n h[n] \cos((n-\alpha)\omega) + \sum_n \underbrace{h[2\alpha - n]}_{= -h[n]} \cos((n-\alpha)\omega) = 0$$

i.e., the $\cos((n-\alpha)\omega)$ terms cancel for each $h[n], h[2\alpha - n]$ pair since $h[2\alpha - n] = -h[n]$.

Symmetry gives generalized linear phase

$h[n] = h[M-n]$ $\quad n = 0, 1, \ldots, M$ $\quad N = M+1 = \#$ of taps

$$H(e^{j\omega}) = \sum_{n=0}^{M} h[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{(M-1)/2} h[n] e^{-j\omega n} + h[M-n] e^{-j\omega(M-n)} + h\left[\tfrac{M}{2}\right] e^{-j\omega M/2}$$

M even, M odd

$h[M-n] = h[n]$

$$= \sum_{n=0}^{(M-1)/2} h[n]\left(e^{-j\omega n} + e^{j\omega n} \cdot e^{-j\omega M}\right) + h\left[\tfrac{M}{2}\right] e^{-j\omega M/2}$$

$$= \left(e^{-j\omega n} e^{j\omega M/2} + e^{j\omega n} e^{-j\omega M/2}\right) e^{-j\omega M/2}$$

$$= 2\cos\left(\omega\left(\tfrac{M}{2} - n\right)\right) e^{-j\omega M/2}$$

$$= \left(\sum_{n=0}^{(M-1)/2} 2h[n] \cos\left(\omega\left(\tfrac{M}{2} - n\right)\right) + h\left[\tfrac{M}{2}\right] \right) e^{-j\omega M/2}$$

$\frac{M}{2} - k \leftarrow n \quad k \leftarrow \frac{M}{2} - n$

$$= \left(\sum_{k=M/2}^{1} 2h\left[\tfrac{M}{2} - k\right] \cos(\omega k) + h\left[\tfrac{M}{2}\right] \right) e^{-j\omega M/2}$$

M even

$\frac{M+1}{2} - k \leftarrow n \quad k \leftarrow \frac{M+1}{2} - n$

$$= \left(\sum_{k=\frac{M+1}{2}}^{1} 2h\left[\tfrac{M+1}{2} - k\right] \cos\left(\omega\left(k - \tfrac{1}{2}\right)\right) \right) e^{-j\omega M/2}$$

M odd

$\sum_{k=M/2}^{1}$ or $\sum_{k=\frac{M-1}{2}}^{0}$

$k \leftarrow \frac{M-1}{2} - n \quad \frac{M-1}{2} - k \leftarrow n$

$\frac{M}{2} - \frac{M+1}{2} + k \quad \frac{M}{2} - \frac{M-1}{2} + k$

$k = \frac{M+1-M-1}{2} \quad \frac{M-1}{2} = \frac{M+1}{2} - k$

Locations of zeros for FIR linear-phase systems

$$H(z) = \sum_{n=0}^{M} h[n] z^{-n}$$

M is order of FIR system

$N = M+1$ taps

$0 \le n \le M$

In the symmetric cases $h[n] = h[M-n]$

$$H(z) = \sum_{n=0}^{M} h[M-n] z^{-n} = \sum_{k=M}^{0} h[k] z^{k} z^{-M} = z^{-M} H(z^{-1})$$

$k \leftarrow M-n$

$M-k \leftarrow n$

$$\Rightarrow \underline{H(z) = z^{-M} H(z^{-1})}$$

types I and II

If z_0 is a zero of $H(z)$, then

$$H(z_0) = z_0^{-M} H(z_0^{-1}) = 0$$

which implies that if $z_0 = re^{j\theta}$ is a zero of $H(z)$, then $z_0^{-1} = \frac{1}{r} e^{-j\theta}$ is also a zero of $H(z)$. When $h[n]$ is real-valued and z_0 is a zero of $H(z)$, then $z_0^* = re^{-j\theta}$ will also be a zero of $H(z)$ and $(z_0^*)^{-1} = \frac{1}{r} e^{j\theta}$ will be a zero as well.

Therefore, when $h[n]$ is real-valued, each complex zero not on the unit circle will be part of a set of four conjugate reciprocal zeros of the form

$$(1 - re^{j\theta} z^{-1})(1 - re^{-j\theta} z^{-1})(1 - \tfrac{1}{r} e^{j\theta} z^{-1})(1 - \tfrac{1}{r} e^{-j\theta} z^{-1})$$

If $H(z)$ has a zero on the unit circle, i.e. $z_0 = e^{j\theta}$, then $z_0^{-1} = e^{-j\theta} = z_0^*$, so zeros on the unit circle come in pairs of the form

$$(1 - e^{j\theta} z^{-1})(1 - e^{-j\theta} z^{-1})$$

If a zero of $H(z)$ is real and not on the unit circle the zeros come in pairs of the form

$$(1 \pm r z^{-1})(1 \pm \tfrac{1}{r} z^{-1})$$

A zero of $H(z)$ at $z=\pm 1$ can appear by itself since $\frac{1}{\pm 1} = \pm 1$ and $(\pm 1)^* = \pm 1$ thus $H(z)$ may have factors of the form

$$(1 \pm z^{-1})$$

A zero at $z=-1$ is particularly important since

$$H(-1) = (-1)^{-M} H(-1)$$

If M is even then $H(-1) = H(-1)$, but if M is odd, then

$H(-1) = -H(-1) \Rightarrow H(-1) = 0$ $H(-1)$ must be $=0$ type II

Thus, for symmetric impulse responses with M odd (type II) the system function must have a zero at $z=-1$ (Nyquist frequency).

For antisymmetric impulse responses $h[n] = -h[M-n]$ $h[M-n] = -h[n]$

$$H(z) = -\sum_{n=0}^{M} h[M-n] z^{-n} = -\sum_{k=M}^{0} h[k] z^{k} z^{-M} = -z^{-M} H(z^{-1})$$

$k \leftarrow M-n$

$M-k \leftarrow n$

$$\Rightarrow H(z) = -z^{-M} H(z^{-1})$$ types III and IV

Complex zeros appear in sets of 4 conjugate reciprocal zeros

$$(1 - re^{j\theta} z^{-1})(1 - re^{-j\theta} z^{-1})\left(1 - \tfrac{1}{r} e^{j\theta} z^{-1}\right)\left(1 - \tfrac{1}{r} e^{-j\theta} z^{-1}\right)$$

For a zero at $z_0 = 1 \Rightarrow H(1) = -H(1) \Rightarrow$ $H(z)$ must have zero at $z=1$ for any M

For a zero at $z_0 = -1 \Rightarrow H(-1) = (-1)^{-(M-1)} H(-1)$

$\Rightarrow$ If M is odd $H(-1) = H(-1)$ no zero required at $z=-1$ type IV

$\Rightarrow$ If M is even $H(-1) = -H(-1)$ which forces $H(-1)=0$ type III